GENERAL CHEMISTRY I: OVERVIEW

Qixi Mi
ShanghaiTech SPST

Blackboard

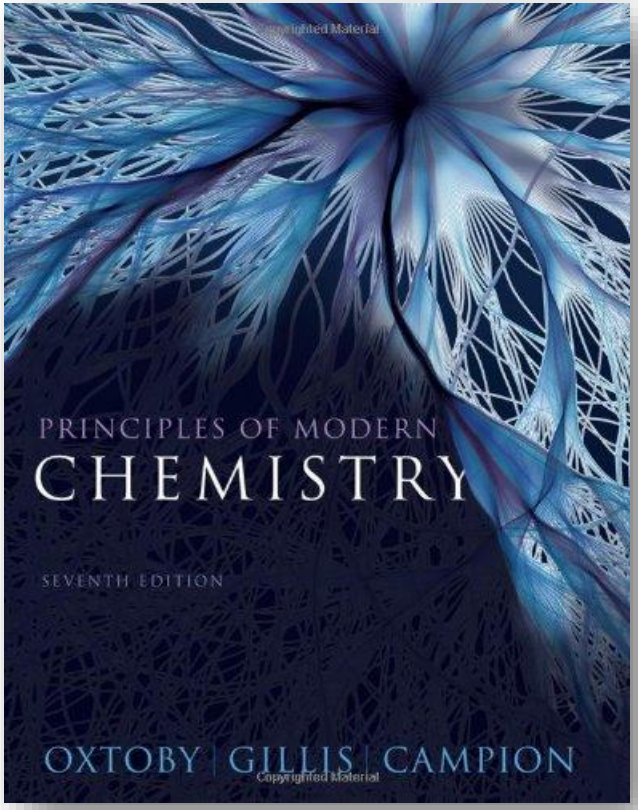
https://elearning.shanghaitech.edu.cn:8443/

Week	Sunday	Monday	Wednesday	Friday
1			**Sept 18** Overview Formula	**Sept 20** Formula Nomenclature Q0
2		**Sept 23** Recitation		**Sept 27** Classical atoms Bonding & valence Q1
3				
4	**Oct 7** PS1 due			**Oct 11** Bohr model Q2
5	**Oct 13** Recitation	**Oct 14** Recitation	**Oct 16** Bohr model Quantum mechanics	**Oct 18** Quantum mechanics Hydrogen atom Q3
6	**Oct 20** PS2 due	**Oct 21** Recitation		**Oct 25** Hydrogen atom Review 1 Q4
		Oct 28	**Oct 30**	**Nov 1**

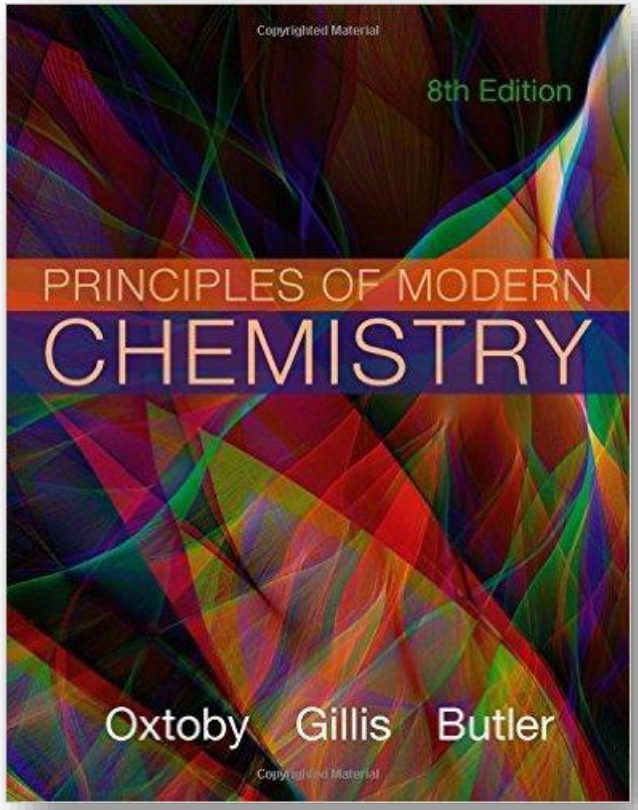
Textbooks

Principles of Modern Chemistry
David W. Oxtoby, H. Pat Gillis, Laurie J. Butler.

7th Edition, 2012
1132 pages
"**OGC7**"

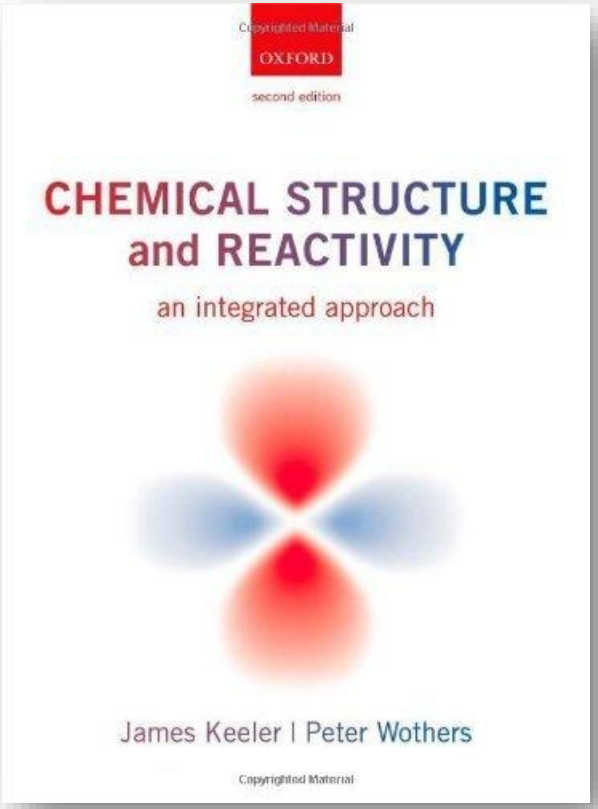


8th Edition, 2016
1136 pages
"**OGB8**"

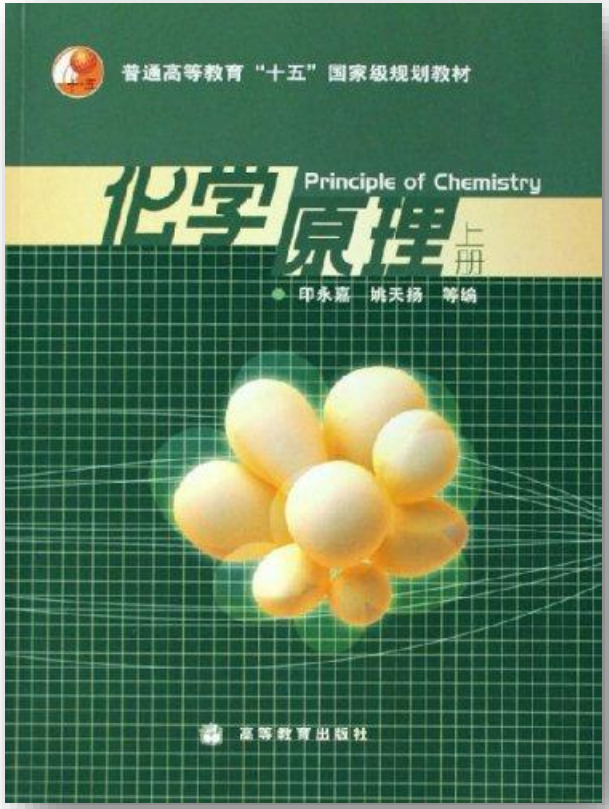


Reference Books

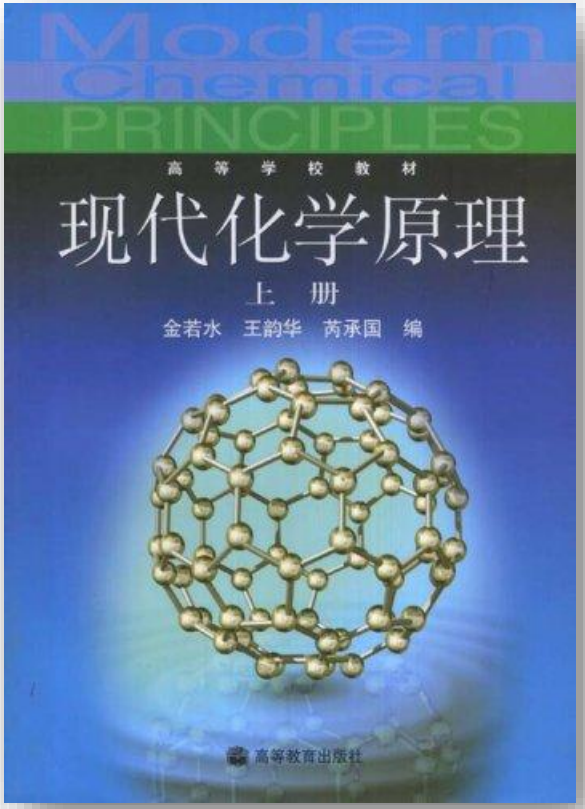
Chemical Structure and Reactivity, 2nd Ed.
J. Keeler, P. Wothers
“**Keeler2**”



《化学原理》
印永嘉，姚天扬
“**YY**”



《现代化学原理》
金若水，王韵华，芮承国
“**JWR**”



Q & A

- Where to get the textbooks?
- How to read the textbooks?
- Can I read the textbooks in Chinese only?

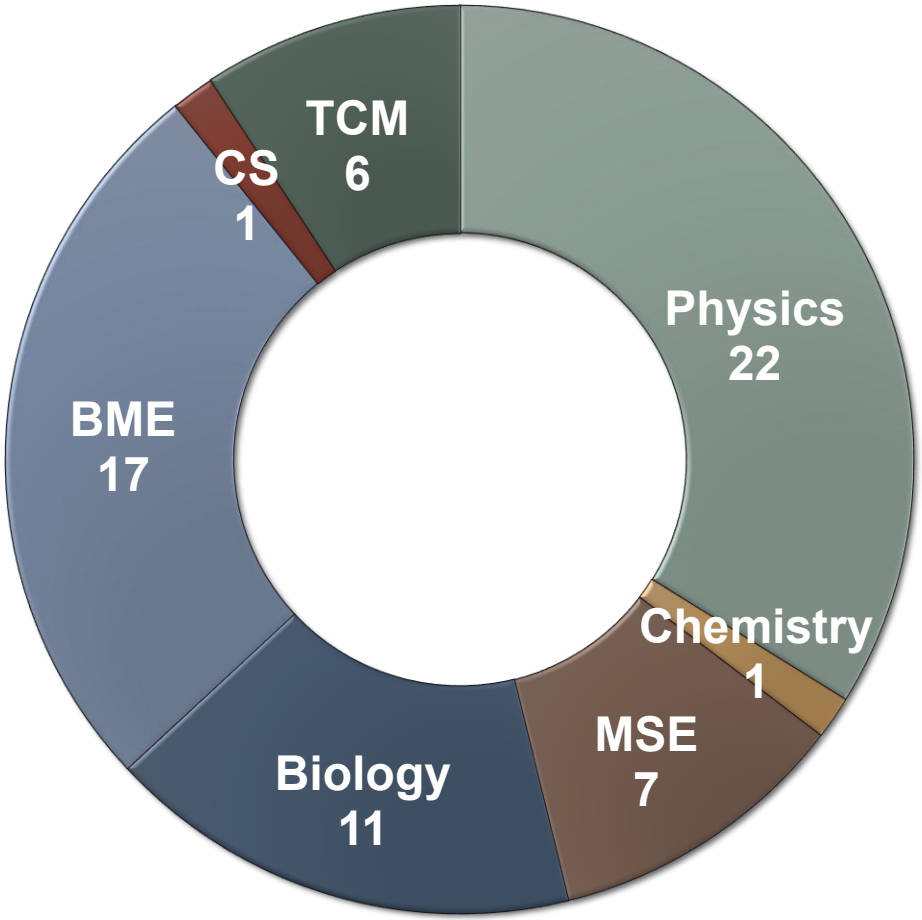
Sessions

	Instructors	Room	Hours	Weeks
Lecture	米启兮	教学中心202	Wed 10:15–11:55 (odd weeks) Fri 10:15–11:55	1–16
Recitation	李常昊	教学中心202	Mon 19:50–20:35	2–16
Office hour	米启兮	物质学院 5号楼205D	Thu 18:30–19:30	2–16
Laboratory	张云艳 等	物质学院2号楼 301/303	Mon/Wed/Thu/Fri Afternoon	2–16

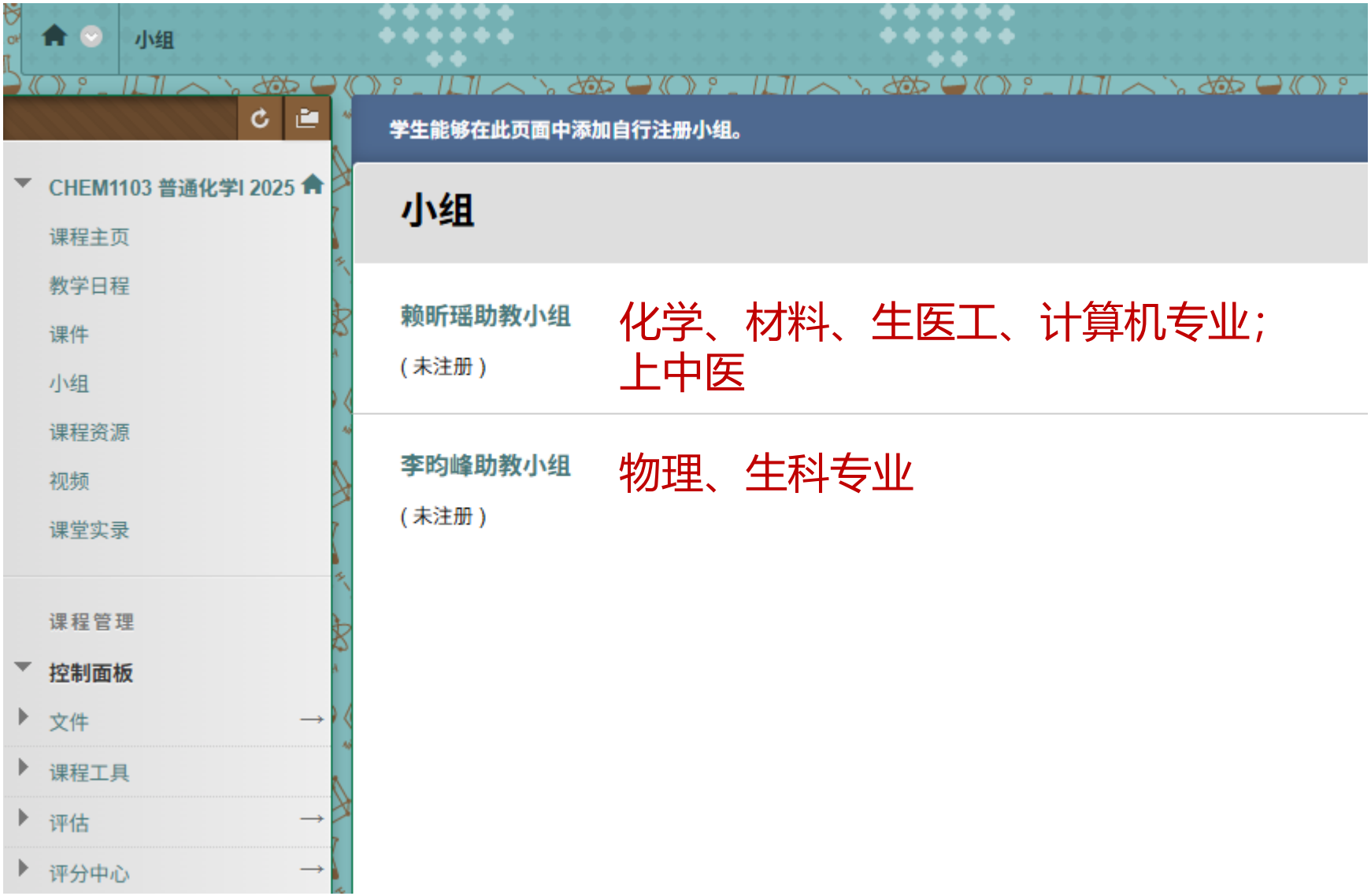
Instructors

岗　位	姓　名	邮　件
理论课	米启兮	miqx@shanghaitech.edu.cn
习题课	李常昊	lichh12025@shanghaitech.edu.cn
习题助教	赖昕瑶	laixy2022@shanghaitech.edu.cn
	李昀峰	liyf22022@shanghaitech.edu.cn

Student Majors



Teaching Assistants



Syllabus

Basics
- Formula
- Classical atom & molecule

Atoms
- Quantum mech.
- Hydrogen atom
- Periodic table

Molecules
- Geometry
- Bonding
- Coordination

Light & Radiation
- Spectroscopy
- *Nuclear chemistry*

OGB8 chapters			
1,2,3	4,5	6,7,8,21	19,20
Keeler2 chapters			
1	2,8,9	4,5,6,10,11,16,17	13,18
YY chapters			
1	2	3,6	

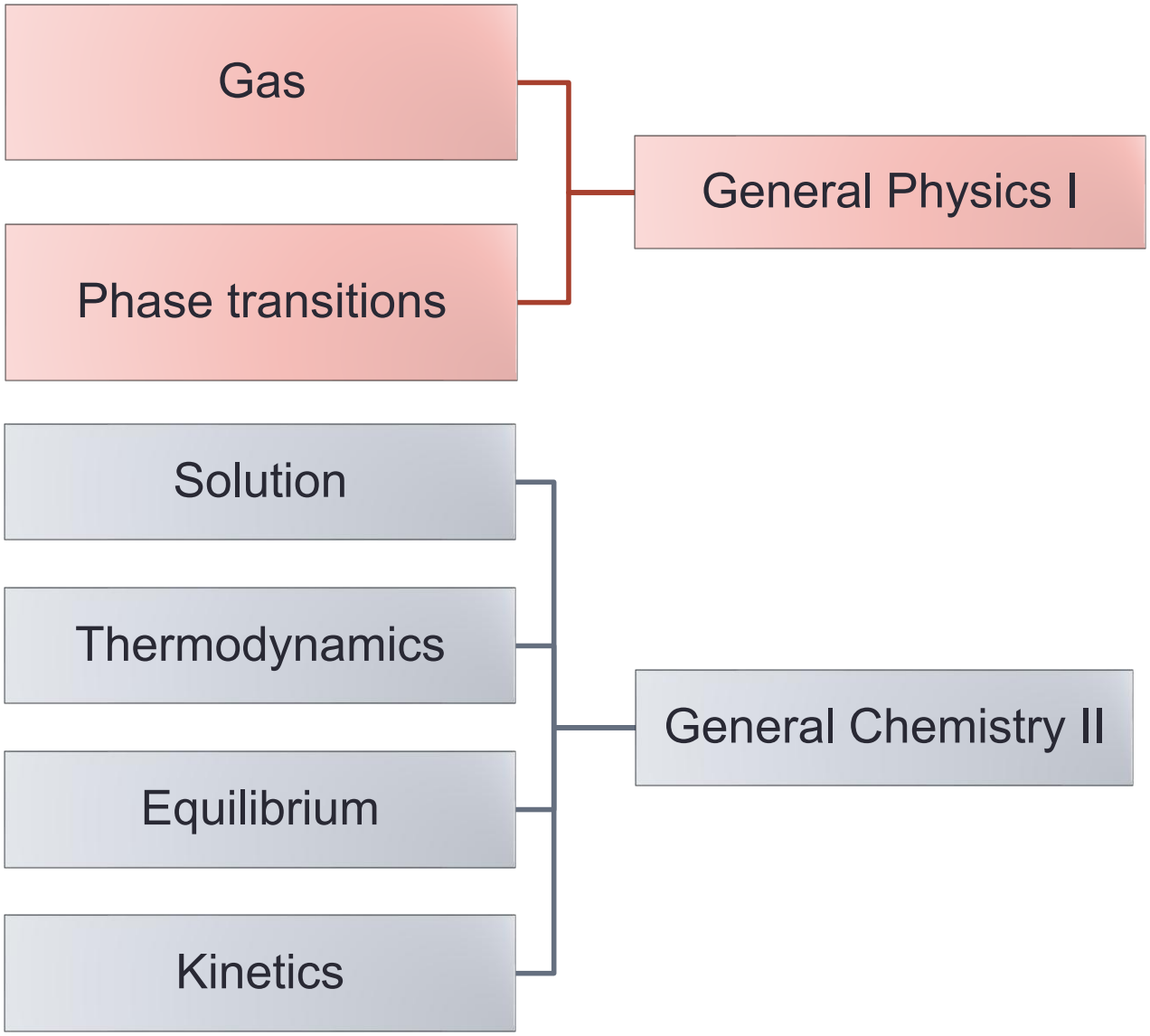
Syllabus (1)

Week	Sunday	Monday	Wednesday	Friday
1			**Sept 17** Overview Formula	**Sept 19** Formula Nomenclature Q0
2		**Sept 22** Recitation		**Sept 26** Classical atoms Bonding & valence Q1
3	**Sept 28** PS1 due			
4				**Oct 10** Bohr model Q2
5		**Oct 13** Recitation	**Oct 15** Bohr model Quantum mechanics	**Oct 17** Quantum mechanics Hydrogen atom Q3
6	**Oct 19** PS2 due	**Oct 20** Recitation		**Oct 24** Hydrogen atom Review 1 Q4
7		**Oct 27** Recitation	**Oct 29** Midterm 1	**Oct 31** Many-electron atoms Q5
8	**Nov 2** PS3 due	**Nov 3** Recitation		**Nov 7** Periodic trends Ionic compounds Q6

Syllabus (2)

Week	Sunday	Monday	Wednesday	Friday
9		**Nov 10** Recitation	**Nov 12** Molecular orbitals	**Nov 14** Diatomic molecules Q7
10	**Nov 16** PS4 due	**Nov 17** Recitation		**Nov 21** Lewis structures Hybridization Q8
11		**Nov 24** Recitation	**Nov 26** VSEPR theory	**Nov 28** Organics: bonding & resonance Q9
12	**Nov 30** PS5 due	**Dec 1** Recitation		**Dec 5** Organics: properties Review 2 Q10
13		**Dec 8** Recitation	**Dec 10** Midterm 2	**Dec 12** Transition metals Metallic bond Q11
14	**Dec 14** PS6 due	**Dec 15** Recitation		**Dec 19** 18-electron rule Q12
15		**Dec 22** Recitation	**Dec 24** Crystal field theory Ligand field theory	**Dec 26** Rare earths Spectroscopy Q13
16	**Dec 28** PS7 due	**Dec 29** Recitation		**Jan 2** Spectroscopy Review 3 Q14

Connections to Other Courses



Evaluation

Quizzes

- Friday
- 2% × (14−1) = 26%

Problem sets

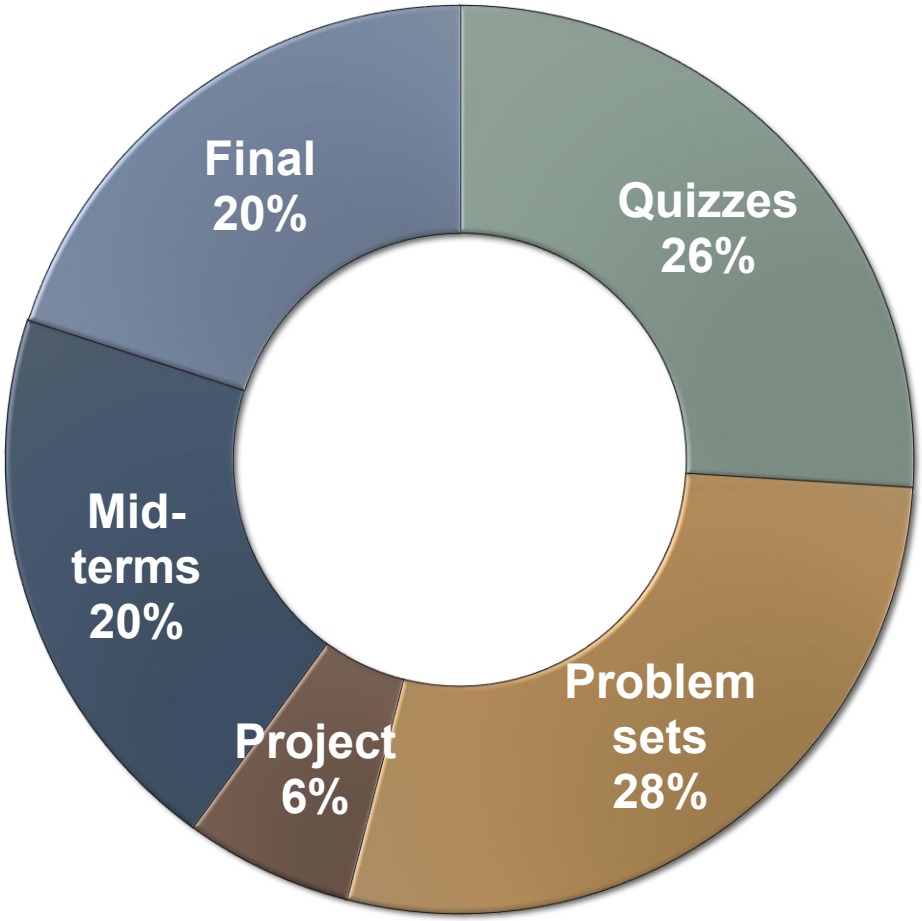
- Due Sunday 24 pm, every other week
- 4% × 7 = 28%

Project 6%

Midterms

- Wed, Oct. 29 and Dec. 10
- 10% × 2 = 20%

Final 20%



Project (1)

金属的气味

小组成员

生命科学与技术学院 吴德育
生命科学与技术学院 吴雨天
生命科学与技术学院 徐牧远

一、背景介绍
1. 物质产生气味的原因
2. 金属的气味（概述）

二、实验现象
1. 气味与金属自身类型的关系
2. 气味与金属所在环境的关系
3. 气味与测试方法的关系

三、分析讨论
1. 金属产生气味的原因
2. 金属气味的意义或潜在应用价值

四、总结

Project (2)

测试信息

描述	模拟考试，不计分数。
说明	
计时的测验	该测试的时间限制为 15 分钟。时间结束时，此测试将自动保存并提交。 剩余**一半时间**、**5 分钟**、**1 分钟**和 **30 秒**时将显示警告。*[预览该测试时不显示计时器]*
多次尝试	此测试允许 3 次尝试。这指尝试次数 1。
强制完成	开始该测试后，必须一次性完成。单击**保存并提交**后才能离开测试。

问题完成状态：

移至另一个问题将保存此回应。

第 **2** 个问题，共 **30** 个问题

问题 2

1 分 保存答案

单词含义

- Radiate
- Explore
- Distill
- Overwhelm

A. v. 压倒
B. v. 探索
C. v. 辐射
D. v. 蒸馏

Cheat Sheet

Periodic Table of the Elements

1 H 1.00794																1 H 1.00794	2 He 4.002602
3 Li 6.941	4 Be 9.012182											5 B 10.811	6 C 12.0107	7 N 14.00674	8 O 15.9994	9 F 18.9984032	10 Ne 20.1797
11 Na 22.989770	12 Mg 24.3050											13 Al 26.981538	14 Si 28.0855	15 P 30.973761	16 S 32.066	17 Cl 35.4527	18 Ar 39.948
19 K 39.0983	20 Ca 40.078	21 Sc 44.955910	22 Ti 47.867	23 V 50.9415	24 Cr 51.9961	25 Mn 54.938049	26 Fe 55.845	27 Co 58.933200	28 Ni 58.6934	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.92160	34 Se 78.96	35 Br 79.904	36 Kr 83.80
37 Rb 85.4678	38 Sr 87.62	39 Y 88.90585	40 Zr 91.224	41 Nb 92.90638	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.90550	46 Pd 106.42	47 Ag 107.8682	48 Cd 112.411	49 In 114.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.60	53 I 126.90447	54 Xe 131.29
55 Cs 132.90545	56 Ba 137.327	57 La 138.9055	72 Hf 178.49	73 Ta 180.9479	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.217	78 Pt 195.078	79 Au 196.96655	80 Hg 200.59	81 Tl 204.3833	82 Pb 207.2	83 Bi 208.98038	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt									

Speed of light
$c = 299\,792\,458\ \mathrm{m \cdot s^{-1}}$

Planck constant
$h = 6.626\,07 \times 10^{-34}\ \mathrm{J \cdot s}$

Vacuum permittivity
$\varepsilon_0 = 8.854\,19 \times 10^{-12}\ \mathrm{F \cdot m^{-1}}$

Elementary charge
$e = 1.602\,18 \times 10^{-19}\ \mathrm{C}$

Electron mass
$m_e = 9.109\,38 \times 10^{-31}\ \mathrm{kg}$

Gas constant
$R = 8.314\,47\ \mathrm{J \cdot mol^{-1} \cdot K^{-1}}$

Avogadro constant
$N_A = 6.022\,14 \times 10^{23}\ \mathrm{mol^{-1}}$

Boltzmann constant
$k_B = 1.380\,65 \times 10^{-23}\ \mathrm{J \cdot K^{-1}}$

Bohr radius
$a_0 = 0.529\,177 \times 10^{-10}\ \mathrm{m}$

Rydberg constant
$R_H = 13.605\,7\ \mathrm{eV}$

Standard atmosphere
$1\ \mathrm{atm} = 101\,325\ \mathrm{Pa}$

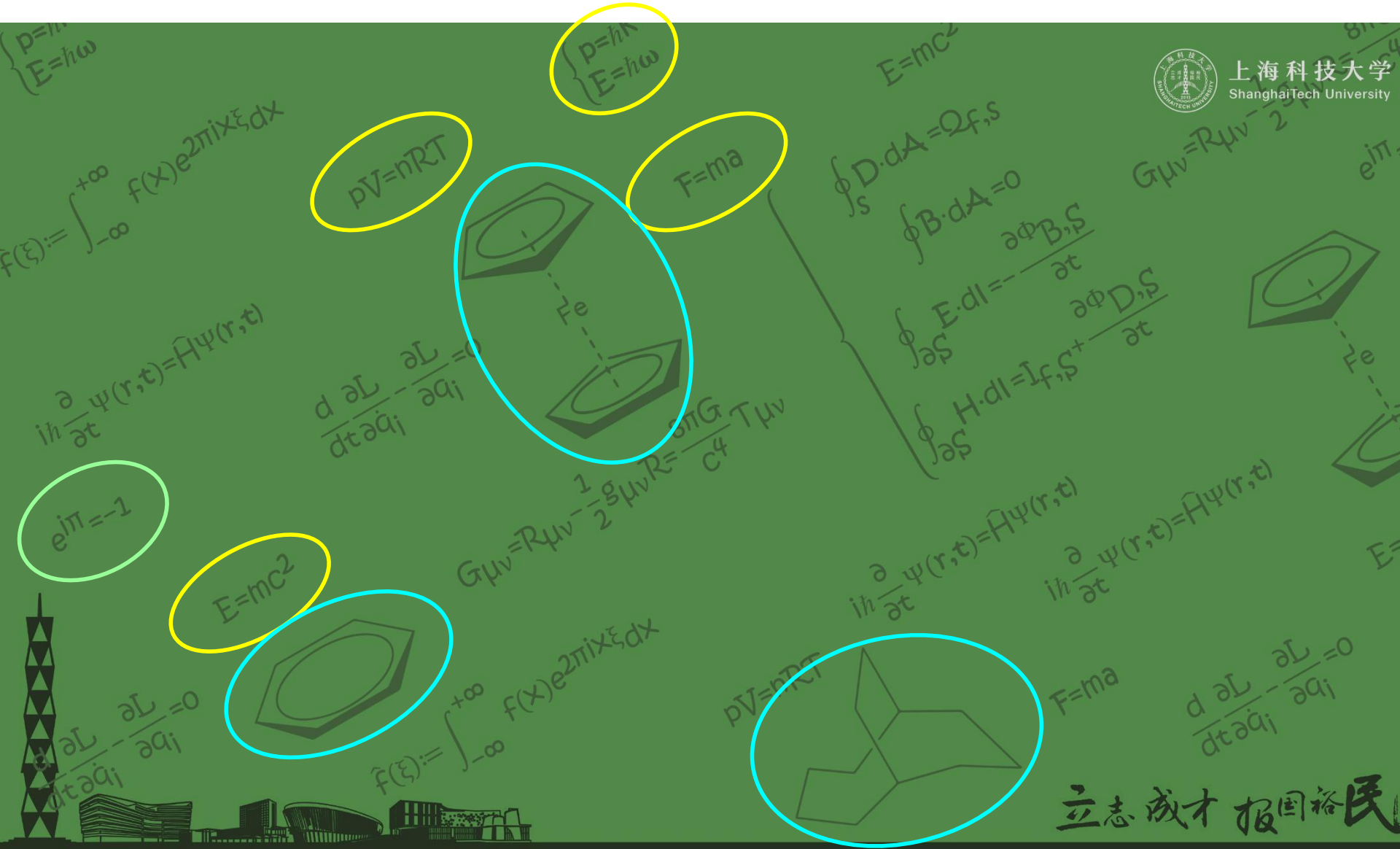
Acceleration of gravity
$g = 9.806\,65\ \mathrm{m \cdot s^{-2}}$

$$r_n = \frac{n^2}{Z} a_0$$

$$E_n = -\frac{Z^2}{n^2} R_H$$

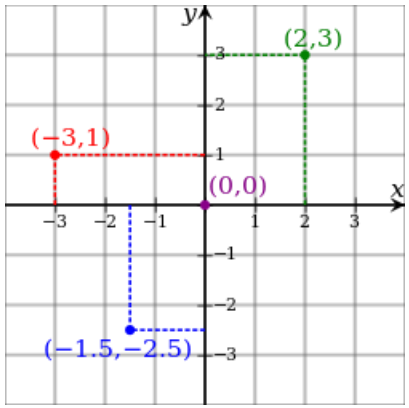
$$L = \sqrt{l(l+1)}\,\frac{h}{2\pi}$$

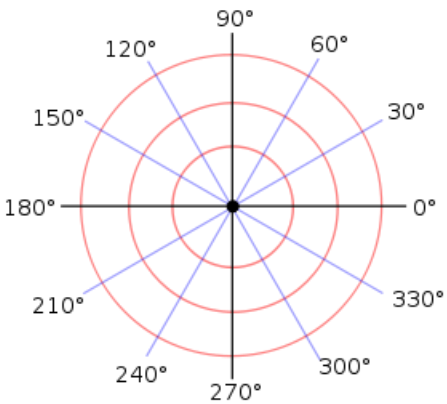
From Beginning to Graduation



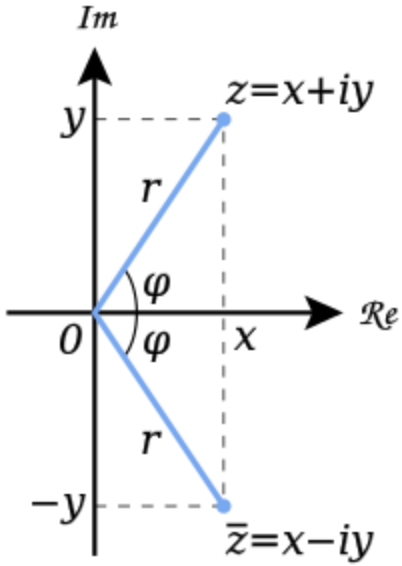
Math Background (1)

Analytic geometry

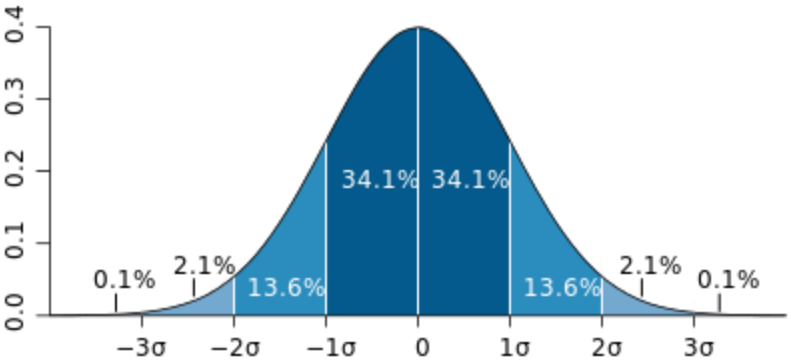


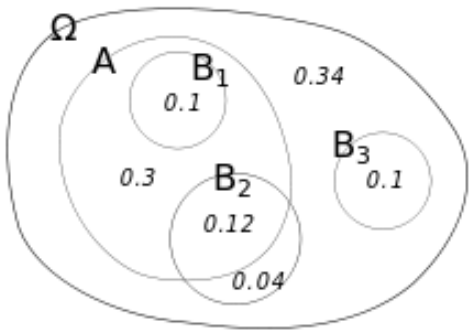


Complex numbers



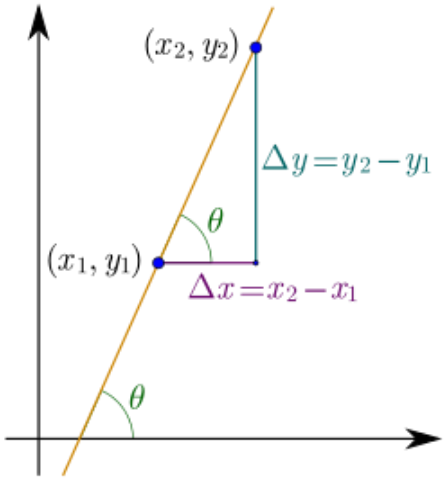
Statistics & probability





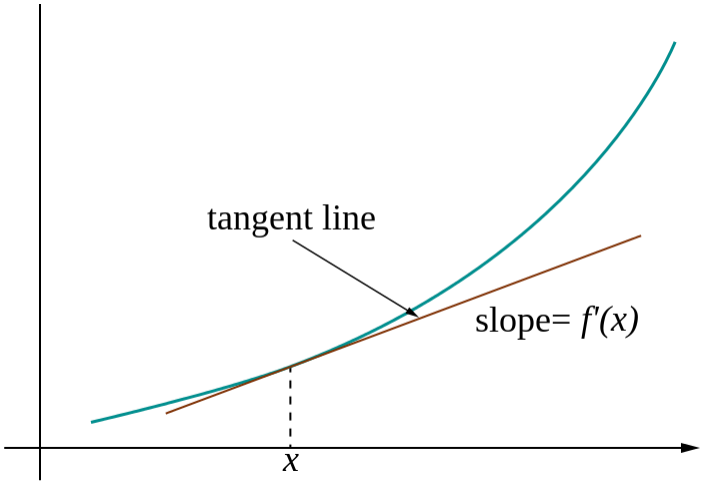
Math Background (2)

Derivative (导数) / differential (微分) calculates a slope (斜率)



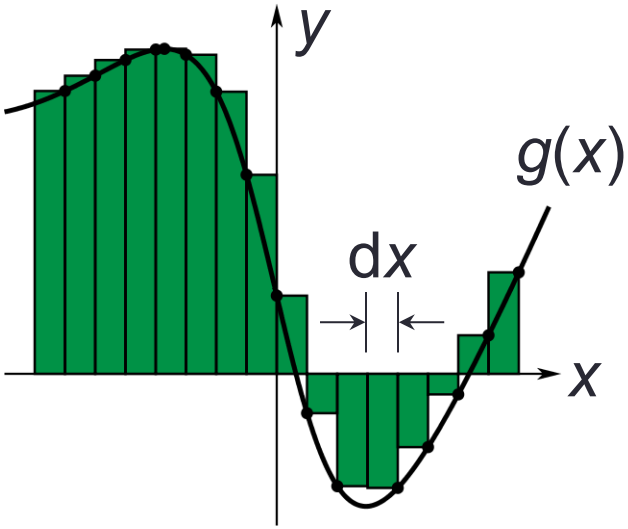
$$k = \frac{\Delta y}{\Delta x}$$

$$f'(x) = \frac{dy}{dx} = \frac{d}{dx}y$$



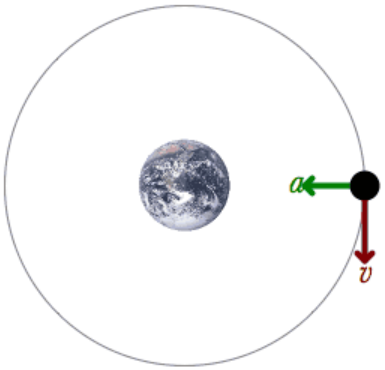
Integration (积分) calculates an area

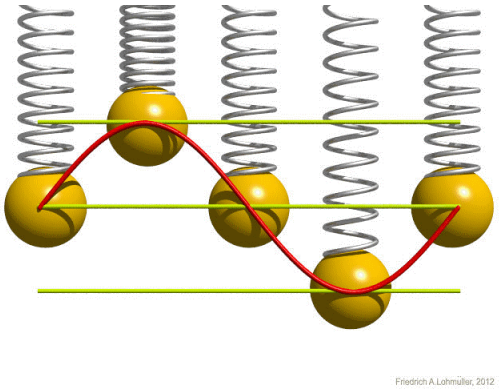
$$A = \int g(x)dx$$



Physics Background

Mechanics





$$E_K = \frac{1}{2}mv^2$$

$$p = mv$$

Electromagnetics: Coulomb's Law

$$F = k\frac{q_1 q_2}{r^2}$$

$$E_p = k\frac{q_1 q_2}{r}$$

What to Learn from This Course?

- Handle vast amounts of (scattered) information
- Identify the key points
- Make imperfect but reasonable judgments

Learn To Make Decisions

水果一斤三块和三斤十块，哪个更便宜?

应该把有限的学习时间重点投入在两三门课上还是雨露均沾?

要水解酯得到羧酸，最好用强酸还是强碱?

哪种方法适合鉴定未知粉末的组成和含量?

各种规则都有例外，我该相信规则还是例外?